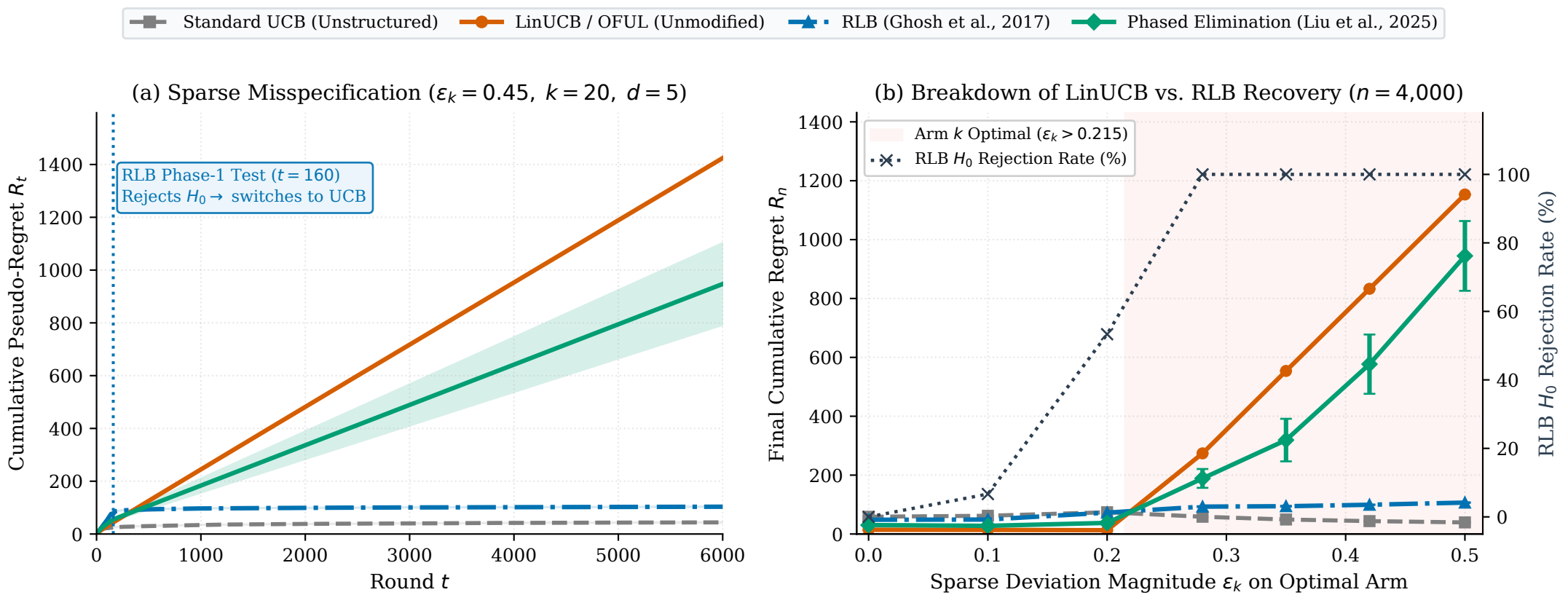
Standard UCB (Unstructured)
LinUCB / OFUL (Unmodified)
RLB (Ghosh et al., 2017)
Phased Elimination (Liu et al., 2025)
(a) Sparse Misspecification ($\varepsilon_k = 0.45,\ k = 20,\ d = 5$)
RLB Phase-1 Test ($t = 160$)
Rejects $H_0 \rightarrow$ switches to UCB
Cumulative Pseudo-Regret R_t
Round t
(b) Breakdown of LinUCB vs. RLB Recovery ($n = 4{,}000$)
Arm k Optimal ($\varepsilon_k > 0.215$)
RLB H_0 Rejection Rate (%)
Final Cumulative Regret R_n
RLB H_0 Rejection Rate (%)
Sparse Deviation Magnitude ε_k on Optimal Arm